

Jay Shah

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Portfolio: <https://jayrshah.vercel.app/> | GitHub: <https://github.com/jays41>

EDUCATION

King's College London

Expected Graduation May 2026

BSc Computer Science (expected 1st Class Honours)

Roundwood Park School

2016 - 2023

A-Levels: Maths (A*), Further Maths (A*), Computer Science (A), Physics (A)

ACHIEVEMENTS

- UKMT Junior Mathematical Olympiad (2018): Distinction and Medal (Top 200 nationally)
- UKMT Intermediate Olympiad - Cayley (2019): Distinction and Medal (Top 100 nationally)

EXPERIENCE

HubSpot

London

Software Engineering Intern

July - September 2025

- Contributed to Data Studio, a flagship HubSpot feature launched at INBOUND 2025 and now used by thousands of customers
- Engineered a smart account verification polling system, reducing recovery time after verification errors from minutes to seconds and eliminating stale verification states
- Built an intelligent sync restart mechanism, reducing redundant API calls and improving sync throughput and reliability
- Strengthened product reliability with expanded automated test coverage, reducing regressions and supporting production releases

Valtech

London

Software Developer Intern (Consulting for Ocado Technologies)

June - July 2024

- Developed and deployed a React component powering rotating product recommendations and advertisements, seen by thousands of Ocado customers daily
- Delivered new frontend features and unit tests (Jest), improving performance and reliability across the platform
- Reviewed peer code contributions and proposed improvements to ensure production quality

PROJECTS

Dissertation: Explainable Regime Discovery in Financial Time Series

- Designed a dual-track XAI framework applying Vector SHAP and TimeSHAP to an unsupervised LSTM autoencoder trained on 25 years of S&P 500 data
- 2-layer stacked LSTM encoder with reverse reconstruction into a 32-dimensional latent space; K-Means with entropy-based model selection and logistic regression soft assignments discovers market regimes unsupervised
- Track 1 attributes cluster membership via exact Vector SHAP; Track 2 attributes reconstruction error to identify structurally anomalous market periods via TimeSHAP; faithfulness validated via feature ablation and sanity checks

Quantitative Trading Hackathon (Quantihack) - 2nd Place (sponsored by Anthropic, Jane Street & Optiver)

- Wheat trading strategy: Generated monthly trading signals from annual supply data using satellite images of wheat farms, achieving 17.5% annualised alpha (Sharpe 0.63, -10.4% max drawdown) over the 12 year backtest
- Built correlation-weighted, z-scored composite signal from lagged features, ranking as the top-performing algorithm for multiple days, finishing 2nd overall in finals

King's Capital (King's Business Club)

Head of Quantitative Trading

Oct 2025 - Present

Quantitative Trading Associate

Oct 2024 - Sept 2025

- Developed and deployed systematic trading strategies across equities and commodities, including market-neutral, factor-tilted, and beta-neutral portfolios
- Built alternative data-driven oil trading strategy using satellite imagery (inventory shadow analysis) and maritime shipping data
- Designed after-hours earnings momentum strategy to capture post-earnings announcement drift
- Implemented volatility arbitrage framework selling ATM straddles, forecasting realised volatility via EGARCH, DS3M, GNN-HAR, and CNN-HAR + Ridge; incorporated earnings blockers and regime filters
- Presented these strategies to Deutsche Bank's Quantitative Investment Solutions team (MDs and VPs), iterating based on their feedback
- Pitched QEPM and volatility strategies to MCD Capital for a potential £100k allocation
- Co-authored a FinBERT sentiment analysis paper and deployed a platform used across the society
- Mentored junior members in Python and algorithmic trading

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C++, Scala, SQL, HTML/CSS

Frameworks/Tools: React, Next.js, Node.js, Django, Flask, Rails, TensorFlow, NumPy, Pandas, Jest, Playwright, Postman, Redis